PROBLEM 50

Double Pendulum

Original statement · Problems.pdf, p. 35

Kinematic Linkage

A double pendulum consists of two heavy point masses P and Q, each of mass m, connected to the origin O and to each other by two rigid, massless bars of length ℓ. We define the inertial frame S_0 with origin O and vector basis $\mathcal{B}_0$. We define the two angles θ and ϕ to describe the configuration of the system.

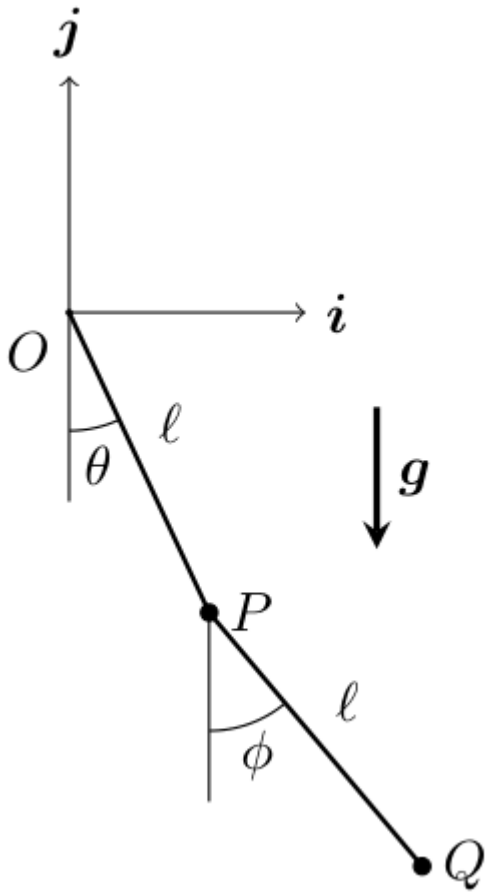


Figure: Double pendulum O–P–Q with bars of length ℓ, angle θ of OP and angle φ of PQ from the vertical (Problems.pdf, p. 35).

(a) Write down the general expressions of the position, velocity, and acceleration vectors of P and Q, using θ, ϕ and their derivatives.

(b) Write down all forces acting on P and Q, introducing new unknowns as needed.

(c) Write down the equations of motion for θ and ϕ, free of other unknowns.

(d) Find and write down one conservation law of this problem.

(e) If the initial conditions are $\theta = \phi = \pi/4, \dot{\theta} = \dot{\phi} = 0$, will Q ever cross the Ox axis? Justify your answer.

(f) Consider now that we impose the additional constraint $\phi = 2\theta$. Compute the natural frequency of oscillation of the system under the assumption of small angles.

(Hint: The mechanical energy of an ideal harmonic oscillator of natural frequency ω_0 and mass m can be written as $\frac{1}{2}m(\dot{x}^2 + \omega_0^2 x^2)$).

Full Analytical Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **Physical System:** Two point masses P and Q, each of mass m, connected by two rigid massless bars of length ℓ.
- First bar connects origin O to P, making angle θ with downward vertical.
- Second bar connects P to Q, making angle ϕ with downward vertical.
- **CDOFs:** 2 degrees of freedom: (θ, ϕ).
- **Parameters:** $m, \ell, g > 0$.

PHASE 2

Reference Frames, Vector Bases & Coordinate Geometry

- Coordinate axes: Ox horizontal to the right, Oz vertically upwards (downward vertical is $-\mathbf{k}$).
- Position of P:

$$\mathbf{r}_P = \ell \sin\theta\, \mathbf{i} - \ell \cos\theta\, \mathbf{k}$$

- Position of Q:

$$\mathbf{r}_Q = \mathbf{r}_P + \ell \sin\phi\, \mathbf{i} - \ell \cos\phi\, \mathbf{k} = \ell(\sin\theta + \sin\phi)\, \mathbf{i} - \ell(\cos\theta + \cos\phi)\, \mathbf{k}$$

PHASE 3

Step-by-Step Mathematical Deduction

Part (a): Velocity and Acceleration Vectors

1. Velocity of P:

$$\mathbf{v}_P = \ell\dot{\theta} \cos\theta\, \mathbf{i} + \ell\dot{\theta} \sin\theta\, \mathbf{k}$$

$$v_P^2 = \ell^2 \dot{\theta}^2$$

2. Velocity of Q:

$$\mathbf{v}_Q = \ell(\dot{\theta} \cos\theta + \dot{\phi} \cos\phi)\, \mathbf{i} + \ell(\dot{\theta} \sin\theta + \dot{\phi} \sin\phi)\, \mathbf{k}$$

$$v_Q^2 = \ell^2 \left[\dot\theta^2 + \dot\phi^2 + 2\dot\theta\dot\phi(\cos\theta\cos\phi + \sin\theta\sin\phi)\right] = \ell^2 \left[\dot\theta^2 + \dot\phi^2 + 2\dot\theta\dot\phi\cos(\theta - \phi)\right]$$

3. Acceleration of P and Q. Define the unit vectors in the $(\mathbf{i}, \mathbf{k})$ plane: $\mathbf{e}_P = \sin\theta\,\mathbf{i} - \cos\theta\,\mathbf{k}$ (from O to P), $\mathbf{e}_\theta = \cos\theta\,\mathbf{i} + \sin\theta\,\mathbf{k}$, $\mathbf{e}_Q = \sin\phi\,\mathbf{i} - \cos\phi\,\mathbf{k}$ (from P to Q), $\mathbf{e}_\phi = \cos\phi\,\mathbf{i} + \sin\phi\,\mathbf{k}$. Their derivatives are $\dot{\mathbf{e}}_P = \dot\theta\,\mathbf{e}_\theta$, $\dot{\mathbf{e}}_\theta = -\dot\theta\,\mathbf{e}_P$ and the same with ϕ. Since $\mathbf{r}_P = \ell\,\mathbf{e}_P$ and $\mathbf{r}_Q = \ell\,\mathbf{e}_P + \ell\,\mathbf{e}_Q$:

$$\mathbf{a}_P = \ell\left(\ddot\theta\,\mathbf{e}_\theta - \dot\theta^2\,\mathbf{e}_P\right) = \ell(\ddot\theta\cos\theta - \dot\theta^2\sin\theta)\,\mathbf{i} + \ell(\ddot\theta\sin\theta + \dot\theta^2\cos\theta)\,\mathbf{k}$$

$$\mathbf{a}_Q = \mathbf{a}_P + \ell\left(\ddot\phi\,\mathbf{e}_\phi - \dot\phi^2\,\mathbf{e}_Q\right)$$

Part (b)-(c): Forces and Equations of Motion (Newton)

> **Pedagogical Justification:** The bars are massless and rigid, so each one transmits a force along its own axis (tension T_1 in OP, T_2 in PQ; a negative value means compression). These two unknowns are eliminated by choosing the projection directions, which is the strategy of Unit 3 for constrained systems.

Forces on P: weight $-mg\,\mathbf{k}$, the pull $-T_1\,\mathbf{e}_P$ of bar OP and the pull $+T_2\,\mathbf{e}_Q$ of bar PQ.
Forces on Q: weight $-mg\,\mathbf{k}$ and the pull $-T_2\,\mathbf{e}_Q$ of bar PQ (action and reaction).

$$m\,\mathbf{a}_P = -mg\,\mathbf{k} - T_1\,\mathbf{e}_P + T_2\,\mathbf{e}_Q, \qquad m\,\mathbf{a}_Q = -mg\,\mathbf{k} - T_2\,\mathbf{e}_Q$$

Scalar products needed ($\mathbf{k}\cdot\mathbf{e}_P = -\cos\theta$, $\mathbf{k}\cdot\mathbf{e}_\theta = \sin\theta$, $\mathbf{k}\cdot\mathbf{e}_Q = -\cos\phi$, $\mathbf{k}\cdot\mathbf{e}_\phi = \sin\phi$):

$$\mathbf{e}_P\cdot\mathbf{e}_Q = \cos(\theta-\phi), \quad \mathbf{e}_\theta\cdot\mathbf{e}_Q = \sin(\phi-\theta), \quad \mathbf{e}_\theta\cdot\mathbf{e}_\phi = \cos(\theta-\phi), \quad \mathbf{e}_P\cdot\mathbf{e}_\phi = \sin(\theta-\phi)$$

Equation for ϕ. Project the equation of Q on $\mathbf{e}_\phi$, which is orthogonal to $\mathbf{e}_Q$ and removes T_2. With $\mathbf{a}_P\cdot\mathbf{e}_\phi = \ell[\ddot\theta\cos(\theta-\phi) - \dot\theta^2\sin(\theta-\phi)]$ and $\ell(\ddot\phi\,\mathbf{e}_\phi - \dot\phi^2\,\mathbf{e}_Q)\cdot\mathbf{e}_\phi = \ell\ddot\phi$:

$$m\ell\left[\ddot{\phi}+\ddot{\theta}\cos(\theta-\phi)-\dot{\theta}^2\sin(\theta-\phi)\right]=-mg\sin\phi$$

$$\ddot{\phi}+\ddot{\theta}\cos(\theta-\phi)-\dot{\theta}^2\sin(\theta-\phi)+\frac{g}{\ell}\sin\phi=0$$

Equation for θ. Add the two equations of motion, so that $T_2\,\mathbf{e}_Q$ cancels, and project on $\mathbf{e}_\theta$, which is orthogonal to $\mathbf{e}_P$ and removes T_1. The right-hand side is $-2mg\,\mathbf{k}\cdot\mathbf{e}_\theta=-2mg\sin\theta$ and the left-hand side is $m(\mathbf{a}_P+\mathbf{a}_Q)\cdot\mathbf{e}_\theta$, with $\mathbf{a}_P\cdot\mathbf{e}_\theta=\ell\ddot{\theta}$ and $\mathbf{a}_Q\cdot\mathbf{e}_\theta=\ell\ddot{\theta}+\ell\ddot{\phi}\cos(\theta-\phi)+\ell\dot{\phi}^2\sin(\theta-\phi)$:

$$2\ddot{\theta}+\ddot{\phi}\cos(\theta-\phi)+\dot{\phi}^2\sin(\theta-\phi)+\frac{2g}{\ell}\sin\theta=0$$

These two second-order equations contain only θ, ϕ and their derivatives, as required in (c). The tensions follow afterwards by projecting on the bar directions: the $\mathbf{e}_Q$ projection of the equation of Q and the $\mathbf{e}_P$ projection of the equation of P give

$$T_2=mg\cos\phi-m\ell\left[\ddot{\theta}\sin(\phi-\theta)-\dot{\theta}^2\cos(\theta-\phi)-\dot{\phi}^2\right],\qquad T_1=mg\cos\theta+m\ell\dot{\theta}^2+T_2\cos(\theta-\phi)$$

Check: at rest at $\theta=\phi=0$ the equations give $T_2=mg$ and $T_1=2mg$, the static tensions of the hanging chain.

Part (d): Conservation Law

The system is conservative with scleronomic constraints. Total mechanical energy is constant:

$$E=\frac{1}{2}m\ell^2\left[2\dot{\theta}^2+\dot{\phi}^2+2\dot{\theta}\dot{\phi}\cos(\theta-\phi)\right]-mg\ell(2\cos\theta+\cos\phi)=\text{constant}$$

Part (e): Will Q Ever Cross the Ox Axis?

- Given initial state: $\theta(0) = \phi(0) = \pi/4$, released from rest ($\dot{\theta}(0) = \dot{\phi}(0) = 0$).
- Initial energy:

$$E_0 = -mg\ell \left(2\cos\frac{\pi}{4} + \cos\frac{\pi}{4}\right) = -3mg\ell\frac{\sqrt{2}}{2} = -\frac{3\sqrt{2}}{2}mg\ell \approx -2.121\, mg\ell$$

- For Q to cross the Ox axis, its vertical coordinate z_Q must reach 0:

$$z_Q = -\ell(\cos\theta + \cos\phi) = 0 \implies \cos\theta + \cos\phi = 0$$

At that instant, the minimum potential energy of the system is:

$$V = -mg\ell(2\cos\theta + \cos\phi) = -mg\ell(\cos\theta + (\cos\theta + \cos\phi)) = -mg\ell\cos\theta$$

Since $\cos\theta \leq 1, V \geq -mg\ell$.

The required energy to cross the axis would be:

$$E_{req} \geq V \geq -mg\ell$$

However, the initial energy is $E_0 = -2.121\, mg\ell < -mg\ell$.

Because $E_0 < E_{req}$, the kinetic energy would have to be negative, which is physically impossible!

Conclusion: Q can **never** cross the Ox axis.

Part (f): Imposed Constraint $\phi = 2\theta$ and Small-Angle Natural Frequency

When $\phi = 2\theta, \dot{\phi} = 2\dot{\theta}$.

For small angles $\theta \ll 1$:

$$\cos(\theta - \phi) = \cos(-\theta) = \cos\theta \approx 1$$

Kinetic energy:

$$T = \frac{1}{2}m\ell^2\left[2\dot{\theta}^2 + (2\dot{\theta})^2 + 2\dot{\theta}(2\dot{\theta})(1)\right] = \frac{1}{2}m\ell^2[2+4+4]\dot{\theta}^2 = 5m\ell^2\dot{\theta}^2$$

Potential energy with $\cos u \approx 1 - \frac{u^2}{2}$:

$$V = -mg\ell(2\cos\theta + \cos(2\theta)) \approx -mg\ell\left[2\left(1 - \frac{\theta^2}{2}\right) + \left(1 - \frac{4\theta^2}{2}\right)\right]$$

$$V \approx -mg\ell[3 - 3\theta^2] = \text{const} + 3mg\ell\theta^2$$

Total energy of the equivalent oscillator:

$$E = 5m\ell^2\dot{\theta}^2 + 3mg\ell\theta^2 = \text{constant}$$

Differentiating with respect to time:

$$10m\ell^2\dot{\theta}\ddot{\theta} + 6mg\ell\theta\dot{\theta} = 0$$

Divide by $10m\ell^2\dot{\theta}$:

$$\ddot{\theta} + \frac{6g}{10\ell}\theta = 0 \implies \ddot{\theta} + \frac{3g}{5\ell}\theta = 0$$

The natural frequency of oscillation is:

$$\omega_0 = \sqrt{\frac{3g}{5\ell}}$$

PHASE 4

Physical Interpretation & Dimensional Verification

- The mechanical coupling $\phi = 2\theta$ acts as an internal kinematic constraint that stiffens the double pendulum into an equivalent single degree of freedom harmonic oscillator with effective mass $10m$ and effective restoring stiffness $6mg/\ell$, yielding the frequency $\omega_0 = \sqrt{3g/(5\ell)}$.
- **Dimensional check:** $[T] = [m\ell^2\dot{\theta}^2] = \mathrm{kg}\cdot\mathrm{m}^2\cdot\mathrm{s}^{-2} = \mathrm{J}$, $[V] = [mg\ell] = \mathrm{J}$, and $[\omega_0] = \sqrt{[g]/[\ell]} = \mathrm{s}^{-1}$ (the effective inertia $10m\ell^2$ and stiffness $6mg\ell$ are expressed in the coordinate θ; with $x = \ell\theta$ they become $10m$ and $6mg/\ell$).
- **Numerical check of part (e) (RK4, step 10^{-3}, $g = 9.81\ \mathrm{m/s^2}$, $m = \ell = 1$, 60 s from $\theta = \phi = \pi/4$ at rest):** the full nonlinear equations of part (c) give a maximum of $z_Q/\ell = -1.3375$, always negative, so Q never reaches the Ox axis, consistent with the energy bound $V \geq -mg\ell$ against $E_0 = -2.121\,mg\ell$.
- **Numerical check of part (f):** the exact one-degree-of-freedom equation of motion with $\phi = 2\theta$ (without the small-angle approximation) integrated by RK4 from $\theta = 10^{-3}$ rad has half-period 1.2950 s, against $\pi/\omega_0 = \pi\sqrt{5\ell/(3g)} = 1.2949$ s.

Source: Problems.pdf (Page 35)